

Infrared and visible image fusion using multi-scale pyramid network

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Deep networks have been widely applied in infrared and visible image fusion. However, the current deep networks cannot well extract and fuse multi-scale information and high-frequency texture features of the source images. In this paper, a deep multi-scale pyramid network, termed MSPFNet, is proposed for infrared and visible image fusion by combining image Laplacian pyramid and deep network. Infrared and visible images are first decomposed into their Laplacian pyramids. For each source image, its Laplacian pyramid consists of a low-frequency component and a series of multi-scale high-frequency components containing texture details. Then, the Laplacian pyramid components of two source images in the same level are fused using convolutional neural networks (CNN). Finally, the final fused image is reconstructed on the fused Laplacian pyramid components using inverse Laplacian pyramid transform. The experimental results on publicly available datasets show that MSPFNet can efficiently extract and fuse the multi-scale detail information of source images, and the fused images of MSPFNet preserve more texture details of infrared and visible images than that of the previous state-of-the-art methods.

Keywords: Convolutional neural networks; Laplacian pyramid; image fusion; infrared and visible.

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1. Introduction

While visible images could capture the texture details of objects in a scene, they usually lose valuable targets due to bad weather. In contrast, infrared images could highlight the target regions in any weather conditions but lose the texture details.^{21,32} Infrared and visible image fusion aims to assemble the contour and texture of the targets in the scene into one robust and informative image, which is helpful for target detection, traffic monitoring, automatic driving, etc.¹¹

The general image fusion methods consist of image feature extraction, feature fusion and image reconstruction. As the diverse implementations, the current infrared and visible image fusion methods can be classified into traditional methods and deep learning-based methods.³⁶ The traditional methods apply hand-crafted algorithms to extract and fuse features, and reconstruct the fused image based on the fused feature, whose final fusion performance heavily depends on the hand-crafted algorithms. Although sophisticated algorithms have been developed, the traditional methods cannot effectively extract the information of source images, leading to information loss in the final fused image.^{20,23} Deep learning-based methods implement all or part of image fusion steps using deep networks, which are well trained on infrared and visible datasets.^{14,15,22,36} The well-trained deep networks can well extract information from the source images, efficiently fuse the extracted features and effectively reconstruct the final fused image.^{19,26,37,42}

Although deep networks have been widely used in infrared and visible image fusion, the current deep networks^{14,22,36} cannot efficiently extract the multi-scale and texture detail information in the source images, which are important for image fusion. The traditional image processing tools, such as Laplacian pyramid transform,² could effectively extract the multi-scale and texture detail information of images. The Laplacian pyramid transform decomposes an image into a low-frequency component and a series of high-frequency components in multi-scale; these components compose the Laplacian pyramid of the image. While the low-frequency component contains the basic information of the original image, the high-frequency components contain its multi-scale and texture detail information.

In this paper, to effectively extract and fuse the multi-scale and texture detail information of source images, we propose a multi-scale pyramid network (MSPFNet) for infrared and visible image fusion. Using the Laplacian pyramid transform, MSPFNet first decomposes infrared and visible images into their Laplacian pyramids. Then, the Laplacian pyramid components of source images in the same level are fused by MSPFNet using convolutional neural networks (CNN). Finally, the final fused image is reconstructed on the fused Laplacian pyramid components using inverse Laplacian pyramid transform. To well train MSPFNet, we design a gradient loss by comparing the multi-scale gradient differences among Gaussian pyramid components of two source images and that of the fused image. As Fig. 1 shows, the fused image of MSPFNet contains rich texture details and highlights the object regions. The main contributions of this paper are summarized

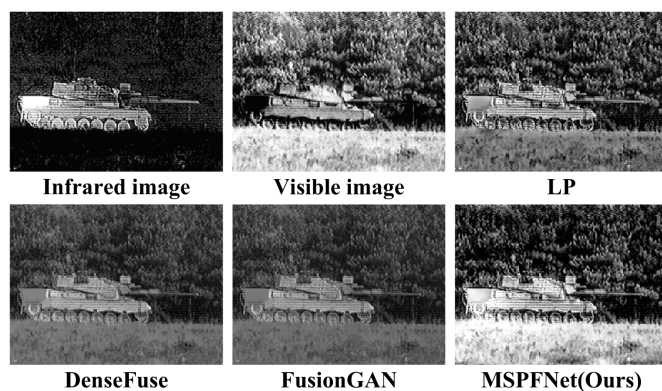


Fig. 1. The fused results of different methods on *tank* images. From top to bottom (from left to right): infrared image, visible image, the fused images of LP,² DenseFuse,¹⁴ FusionGAN²² and our MSPFNet. While the fused images of LP, DenseFuse and FusionGAN contain blurred texture and lose target energy, the fused image of MSPFNet contains rich texture details and highlights the “tank” region.

as follows:

- (1) We design a multi-scale pyramid image fusion network (MSPFNet) for infrared and visible image fusion, combining image Laplacian pyramid transform and CNN.
- (2) A multi-scale gradient loss is designed by comparing the multi-scale gradient difference among Gaussian pyramid components of two source images and that of the fused image, which is used to train MSPFNet to extract and fuse the multi-scale information of source images.
- (3) Compared with the existing image fusion methods, MSPFNet performs better in terms of both subjective and objective evaluation.

2. Related Work

2.1. Traditional method

Traditional image fusion methods include multi-scale transform-based methods, compressed sensing and sparse representation-based methods, etc.¹¹

Multi-scale transform-based methods first decompose the source images into a series of sub-images in different scales, then fuse the sub-images using proper rules, and finally reconstruct the fused image using corresponding inverse transform on the fused sub-images.^{2,4,16,25} Multi-scale transform-based methods are usually with low complexity. Laplacian pyramid method (LP)² decomposed the source images into multi-scale pyramid sequences and fused infrared and visible images in different scales. Multi-resolution singular value decomposition method (MSVD)²⁵ used singular value decomposition to replace the finite impulse response filter in

wavelet transform,³ and the method had high computational efficiency and met real-time requirements. Zhang *et al.* decomposed source images using nonsubsampling contourlet transform for image fusion, and they designed a high-frequency sub-band fusion rule to improve fusion performance.⁴⁰ MDLatLRR method decomposed images into multi-level sub-images based on the latent low-rank representation.¹⁶ Rolling guidance filter (RGF)⁴¹ and nonsubsampling shearlet transform (NSST)²⁴ are also commonly used multi-scale transform methods. Zhao *et al.* decomposed source images into base layer and detail layers using RGF; then, the base layers were fused by an improved saliency detection method, and the detail layers were fused by directional gradient sum and adaptive double-channel spiking cortical model.⁴⁴ Zhao *et al.* decomposed source images into base layer and detail layers using RGF, and then the base layer was decomposed into high-frequency coefficients and low-frequency coefficients using NSST; secondly, the low-frequency coefficients were fused by visual saliency map, the high-frequency coefficients were fused by gradient domain guided filtering and improved Laplacian sum, and the detail layers were fused by phase congruency and gradient domain guided filtering.⁴³ These two methods can well extract the target information of infrared images and the background texture details of visible images.

Besides multi-scale transform, compressed sensing and sparse representation are also applied to image fusion.^{18,33} Liu *et al.* proposed an infrared and visible image fusion method based on compressed sensing, which could effectively preserve the contour information and texture details of source images and reduce computational complexity.¹⁸ Wang *et al.* used non-negative sparse representation to extract the features of source images and designed fusion rules based on the activity and sparseness levels of non-negative sparse representation coefficients.³³

2.2. Deep learning-based method

In recent years, deep learning-based image fusion methods have gained extensive attention. Deep networks adaptively learn the intrinsic connection between inputs and outputs in a data-driven way and perform excellently in computer vision, speech recognition and natural language processing.¹ The common deep networks, such as CNN,¹³ deep autoencoder (DAE)¹⁰ and generative adversarial network (GAN),⁷ significantly improve the performance of target detection, image recognition, image restoration and image fusion, etc.

Deep networks can be combined with traditional methods to fuse images. In this case, feature extraction, feature fusion, or image reconstruction are implemented using deep networks.^{14,15} DenseFuse¹⁴ fused images using a well-trained DAE. In DenseFuse, while the encoder and decoder were used for feature extraction and image reconstruction, respectively, a hand-crafted fusion rule was applied to fuse the features. Li *et al.* proposed an infrared and visible image fusion method using a deep learning framework (IVIFDLF).¹⁵ IVIFDLF first decomposed the source

images into base and detail parts using guided filtering, then extracted the deep feature information of the detail part using well-trained VGG19,³¹ and the features were fused according to the richness of the deep feature information.

Deep networks can directly fuse infrared and visible images.^{22,23,36} FusionGAN²² was the first GAN for image fusion. While the generator output fused images, the discriminator measured the differences between fused images and visible images to preserve more texture details in the final fused image. U2Fusion³⁶ measured the importance of source images on the gradient information of features extracted by VGG16³¹ and designed a loss function using the importance of source images to make the fused image contain richer texture details. GANMcC²³ designed a multi-classification discriminator, which simultaneously estimated the differences between the fused images and two source images. SDNet³⁹ transformed multiple fusion problems into the extraction and reconstruction of gradient and intensity information and proposed a versatile squeeze-and-decomposition network to realize multi-modal and digital photography image fusion. RFN-Nest¹⁷ was a two-stage training network for image fusion. RFN-Nest first trained an autoencoder based on the nest connection concept and then trained the residual fusion network by well-designed loss functions.

2.3. Laplacian pyramid

The existing deep learning-based methods cannot effectively utilize the multi-scale information and high-frequency texture features of infrared and visible images, resulting in texture detail information losing in the fused images. As a commonly used multi-scale decomposition tool, Laplacian pyramid transform could decompose an image into its Laplacian pyramid, which contains a low-frequency component and a series of high-frequency components in multi-scale.^{34,35} Figure 2 shows the flow chart of Laplacian pyramid decomposition and reconstruction for an example image I .

Laplacian pyramid transform of an image I consists of two steps. The image I is first decomposed into Gaussian pyramid sequence $G(I) = \{G_i(I), i = 1, 2, \dots, N\}$ using Gaussian filtering and downsampling, where

$$G_i(I) = \begin{cases} I, & i = 1, \\ \mathbb{D}(\mathbb{G}(G_{i-1}(I))), & i = 2, \dots, N. \end{cases} \quad (2.1)$$

In Eq. (2.1), $G_i(I)$ represents the i th sub-image in the Gaussian pyramid. $\mathbb{G}$ represents the Gaussian filtering operation and $\mathbb{D}$ represents downsampling operation, i.e.

$$\mathbb{D}(I)(x, y) = I(2x - 1, 2y - 1),$$

where (x, y) represents coordination in the image.

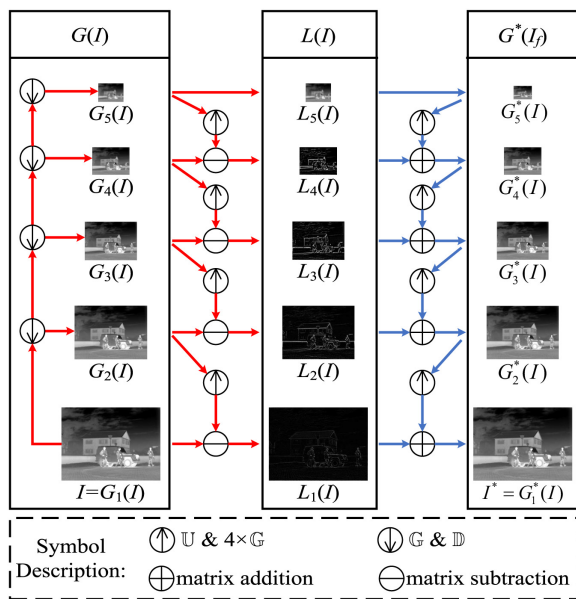


Fig. 2. The flow chart of a 5-layer Laplacian pyramid transform for an example image I .

Then, the Laplacian pyramid is calculated on Gaussian pyramid,

$$L_i(I) = \begin{cases} G_i(I), & i = N, \\ G_i(I) - 4 \times \mathbb{G}(\mathbb{U}(G_{i+1}(I))), & i = N-1, \dots, 1. \end{cases} \quad (2.2)$$

In Eq. (2.2), $\mathbb{U}$ denotes the upsampling operation,

$$\mathbb{U}(I)(x, y) = \begin{cases} I\left(\frac{x+1}{2}, \frac{y+1}{2}\right), & \frac{x+1}{2}, \frac{y+1}{2} \in \mathbb{N}^+, \\ 0, & \text{else.} \end{cases}$$

$\mathbb{N}^+$ represents the set of positive integers.

The reconstructed image is obtained by Inverse Laplacian pyramid transform according to Eq. (2.3).

$$G_i^*(I) = \begin{cases} L_i(I), & i = N, \\ L_i(I) + 4 \times \mathbb{G}(\mathbb{U}(G_{i+1}^*(I))), & i = N-1, \dots, 1, \end{cases} \quad (2.3)$$

where the final reconstructed image $I^* = G_1^*(I)$.

3. Proposed Method

In order to effectively extract multi-scale information and high-frequency texture features of infrared and visible images, this paper proposes a multi-scale pyramid deep network (MSPFNet) for infrared and visible image fusion by combining image Laplacian pyramid transform and CNN.

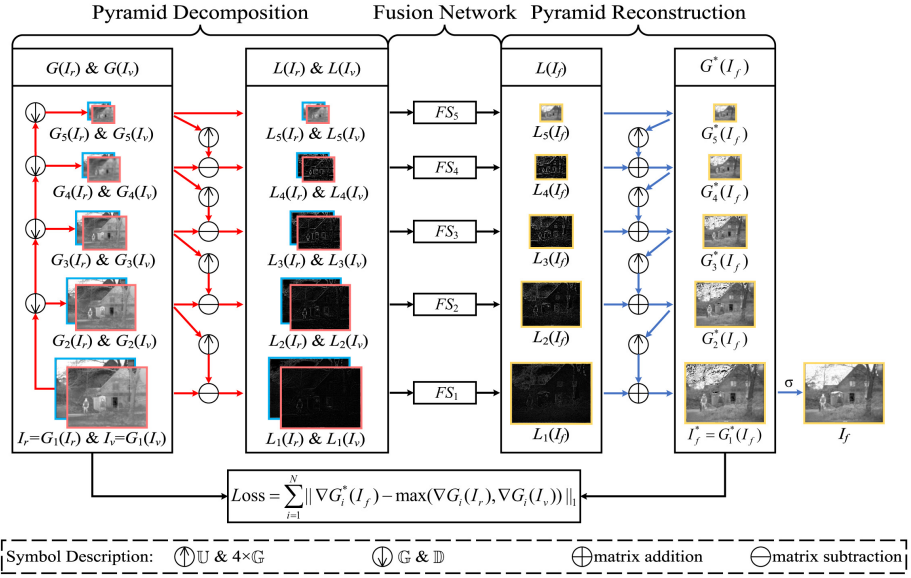


Fig. 3. The pipeline of MSPFNet ($N = 5$). MSPFNet includes three parts: Pyramid Decomposition, Fusion Network and Pyramid Reconstruction. Pyramid Decomposition first decomposes infrared image I_r and visible image I_v into their Gaussian pyramids $G(I_r)$, $G(I_v)$, and their Laplacian pyramids $L(I_r)$, $L(I_v)$. Then, Fusion Network $\{FS_i, i = 1, 2, \dots, N\}$ fuses the Laplacian pyramid components of two source images in the same level and output the fused Laplacian pyramid components $L(I_f)$. Finally, Pyramid Reconstruction reconstructs the final fused image I_f on $L(I_f)$ using inverse Laplacian pyramid transform and normalized operation σ .

3.1. MSPFNet pipeline

Figure 3 illustrates MSPFNet, which includes three steps: Pyramid Decomposition step, Fusion Network step and Pyramid Reconstruction step.

According to Eq. (2.2), Pyramid Decomposition step first decomposes infrared image I_r and visible image I_v into their Gaussian pyramids $G(I_r) = \{G_i(I_r), i = 1, 2, \dots, N\}$ and $G(I_v) = \{G_i(I_v), i = 1, 2, \dots, N\}$, then calculates their Laplacian pyramids by Eq. (2.3), $L(I_r) = \{L_i(I_r), i = 1, 2, \dots, N\}$, $L(I_v) = \{L_i(I_v), i = 1, 2, \dots, N\}$.

Fusion Network contains N -layer Fusion Subnets $\{FS_i, i = 1, 2, \dots, N\}$. Fusion Subnets fuse Laplacian pyramid sequences $L(I_r)$ and $L(I_v)$ to obtain the fused Laplacian pyramid components $L(I_f) = \{L_i(I_f), i = 1, 2, \dots, N\}$, where $\{L_i(I_f) = FS_i(L_i(I_r), L_i(I_v)), i = 1, 2, \dots, N\}$.

Fusion Subnets $\{FS_i, i = 1, 2, \dots, N\}$ share their parameters, i.e. $FS_1 = FS_2 = \dots = FS_N$, which reduces the network complexity.

Pyramid Reconstruction step contains the inverse Laplacian pyramid transform, which reconstructs fused image I_f^* on the fused Laplacian pyramid components $L(I_f) = \{L_i(I_f), i = 1, 2, \dots, N\}$, and the final fused image I_f is achieved by normalized operation σ , $I_f = \sigma(I_f^*)$.

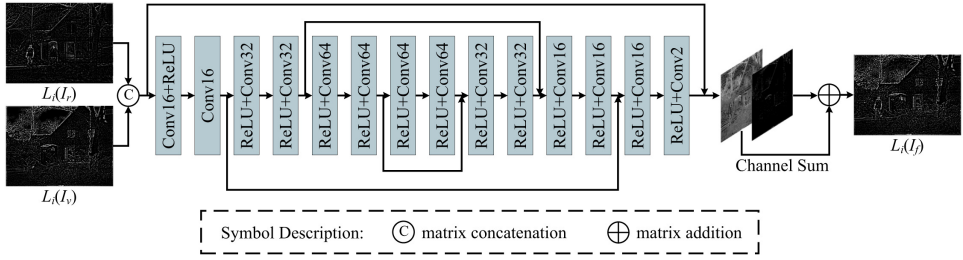


Fig. 4. The architecture of Fusion Subnet. For each Fusion Subnet FS_i , it takes the i th Laplacian pyramid component of infrared image $L_i(I_r)$ and that of visible image $L_i(I_v)$ as inputs, whose output, $L_i(I_f)$, is taken as the i th fused Laplacian pyramid component.

3.2. Fusion network architecture

Since Fusion Subnets share their parameters, Fig. 4 only shows the architecture of one layer of the Fusion Subnets, which contains 14 convolutional layers and four skip connections among them. Skip connections help to extract source image information and avoid exploding/vanishing gradients.^{9,30} The convolutional filters are with size of 3×3 , stride of 1 and padding of 1. In Fig. 4, $\text{Conv}k$ represents the convolutional operation, where k is its output channel size. Every convolutional layer is followed by a Rectified Linear Unit (ReLU) except the last one.

The two-channel output of the last convolutional layer is summed along the channel direction to obtain a single-channel output, and this operation is called Channel Sum. Channel Sum helps the final fused images to contain more texture details and high-contrast information. In Sec. 4.5.2, an ablation experiment is performed to demonstrate the effect of Channel Sum.

3.3. Multi-scale gradient loss function

In order to extract more texture details from source images and fuse them into the final fused image, we design a multi-scale gradient loss L_{msg} by calculating the loss between the gradient of the Gaussian pyramid components of the fused image and the maximum value of the gradients of the Gaussian pyramid components of two source images at every layer:

$$L_{\text{msg}} = \sum_{i=1}^N \|\nabla G_i^*(I_f) - \max(\nabla G_i(I_r), \nabla G_i(I_v))\|_1, \quad (3.1)$$

where ∇ denotes the gradient operation.

4. Experiments

4.1. Experimental settings

In the experiments, we set the image Laplacian pyramid level number as five, i.e. $N = 5$. We use Adam optimizer,¹² and the learning rate is set as $10e^{-4}$. The

Epoch is set as 20, and the Batch Size is 20. Xavier⁶ is used to initialize network parameters. MSPFNet is implemented in the Pytorch framework and trained on an NVIDIA GeForce RTX 2080Ti GPU.

In the experiments, 38 pairs of infrared and visible images are selected from the *TNO* dataset^a to produce the training dataset. These images are cropped to 13355 image patch pairs with size of 200×200 , and stride of 20.

During the test phase, three datasets are chosen to validate the superiority of MSPFNet: (1) 16 pairs of infrared and visible images selected from *TNO* dataset; (2) 32 pairs of infrared and visible images selected from *Nato_camp_sequence* in *TNO* dataset; (3) 23 pairs of infrared and visible images selected from *RoadScene*³⁶ dataset.^b

We compare MSPFNet with seven state-of-the-art methods, including three traditional methods (MSVD,²⁵ MDLatLRR,¹⁶ LP²) and four deep learning-based methods (DenseFuse,¹⁴ FusionGAN,²² RFN-Nest,¹⁷ GANMcC²³).

Six metrics are used for quantitative comparisons, including EN,²⁹ SD,²⁸ SF,⁵ MI,²⁷ $Q^{AB/F}$ ³⁸ and VIF.⁸ EN measures the amount of information contained in an image, and the larger EN means the image contains more information. SD measures the contrast of one image, the larger SD means the image contains more texture details and the targets are more prominent. SF measures the spatial frequency information contained in an image, and the larger SF means the image contains more texture details. MI measures the amount of information transferred from source images to the fused images, and the larger MI means the fused image preserves more information of source images. $Q^{AB/F}$ measures the amount of edge information transferred from source images to the fused image, and the larger $Q^{AB/F}$ means the fused image preserves more edge information of source images. VIF measures whether the fused image is consistent with the human visual system, and the larger VIF means the fused image is more realistic and natural.

4.2. *TNO*

Table 1 shows the mean quantitative results of MSPFNet and the seven methods on *TNO* dataset. In the table, the optimal values are highlighted in bold red font, and the suboptimal values are highlighted in bold blue font. In Table 1, MSPFNet achieves three optimal values (EN, MI, VIF) and three suboptimal values (SD, SF, $Q^{AB/F}$), which illustrates that MSPFNet can extract more information (EN, MI) and its fused images are with richer texture details (SD, SF, $Q^{AB/F}$) and more realistic visual effects (VIF).

Figure 5 shows the qualitative results of MSPFNet and the seven methods in *airplane_in_trees*, *sandpath* and *Kaptein_1123*. In the fused images of MSVD, MDLatLRR, DenseFuse, FusionGAN, RFN-Nest and GANMcC, the texture details

^a<http://opticalengineering.spiedigitallibrary.org/article.aspx?articleid=22177>.

^b<https://github.com/jiayi-ma/RoadScene>.

Table 1. Mean quantitative results for fused images on *TNO* dataset. (Red: optimal, Blue: suboptimal)

Method	EN	SD	SF	MI	$Q^{AB/F}$	VIF
MSVD ²⁵	6.3173	8.4958	5.6116	1.9664	0.3181	0.3086
MDLatLRR ¹⁶	6.3216	8.4755	4.4345	2.1477	0.3958	0.3247
LP ²	6.6445	8.6477	6.2225	1.7220	0.5617	0.3918
DenseFuse ¹⁴	6.2736	8.4600	4.0911	2.3360	0.2922	0.3080
FusionGAN ²²	6.4777	8.2545	4.0523	2.2246	0.1914	0.3206
RFN-Nest ¹⁷	6.8105	8.9791	3.9334	2.1563	0.3026	0.3771
GANMcC ²³	6.6399	9.0170	4.1091	2.2482	0.2640	0.3343
MSPFNet	6.9736	8.9833	6.1277	2.4344	0.5522	0.4429

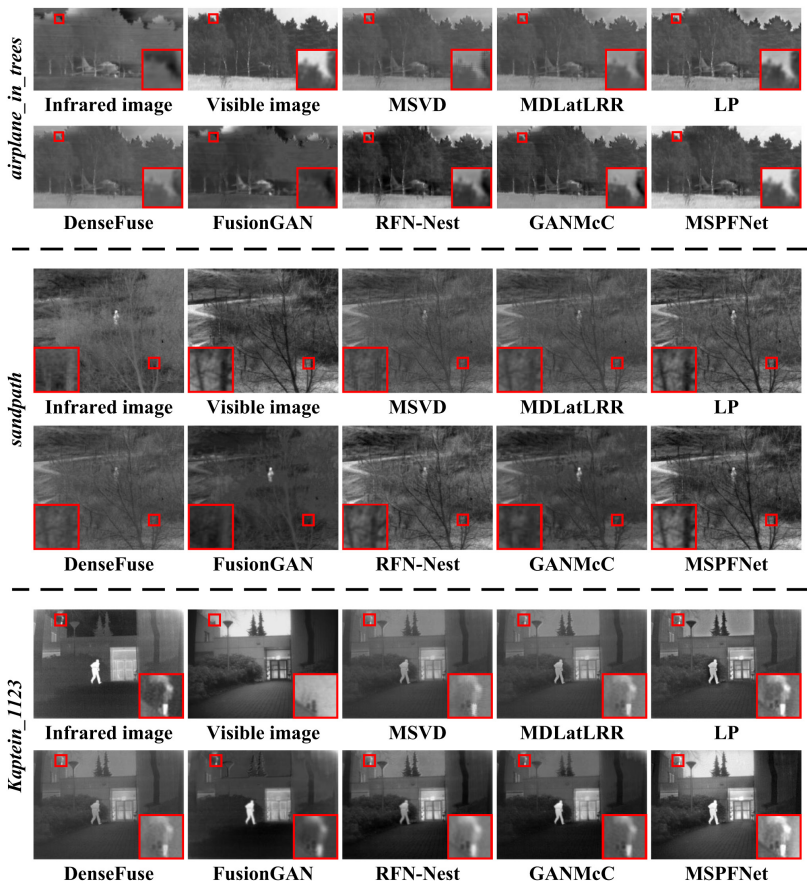


Fig. 5. Qualitative results of MSPFNet and seven comparison methods in three pairs of infrared and visible images from *TNO* dataset. From top to bottom (from left to right): infrared image, visible image, the fused results of MSVD, MDLatLRR, LP, DenseFuse, FusionGAN, RFN-Nest, GANMcC and our MSPFNet.

are blurred and the structural information is lost. In contrast, the fused images of MSPFNet preserve more target intensity information of infrared images and retain more texture details of visible images, which contain clearer texture details, more complete structures, and are more consistent with the human visual system. As Fig. 5 shows, some local regions in the fused images are marked using red boxes, and they are enlarged for better illustration. Taking image *airplane_in_trees* for example, in their marked “tree” regions, MSPFNet has clearer texture details and more complete edge structures, while the other methods produce artifacts on the tree edges and cannot show the climate information of the “sky” regions under natural conditions.

4.3. *Nato_camp_sequence*

Table 2 shows the mean quantitative results of MSPFNet and the seven methods on *Nato_camp_sequence* dataset. From the table, we can observe that MSPFNet achieves three optimal values (EN, $Q^{AB/F}$, VIF) and one suboptimal value (SF), which indicates that MSPFNet can preserve more information (EN), and its fused images contain richer texture details (SF, $Q^{AB/F}$) and more realistic visual effects (VIF). Although FusionGAN is better than MSPFNet on SD and MI, it distorts texture details in its fused images. LP achieves optimal value on SF, but has more noises in the fused images than MSPFNet. For MI, FusionGAN and GANMcC achieve the optimal value and suboptimal value, respectively, but the texture details of their fused images are blurry. In summary, our proposed MSPFNet has the best overall performance.

Figure 6 shows the qualitative results of MSPFNet and the seven methods in a sample image on *Nato_camp_sequence* dataset. Compared with the other seven methods, MSPFNet has clearer texture, higher contrast, and more prominent targets in its fused image. As Fig. 6 shows, the “human” regions in the fused images are marked using green boxes and the “tree” regions are marked using red boxes. From the marked “human” regions, one can find that MSPFNet can effectively highlight the intensity information while that of MSVD, MDLatLRR, DenseFuse

Table 2. Mean quantitative results for fused images on *Nato_camp_sequence* dataset. (Red: optimal, Blue: suboptimal)

Method	EN	SD	SF	MI	$Q^{AB/F}$	VIF
MSVD ²⁵	6.2685	7.9205	6.1440	1.4940	0.2832	0.2688
MDLatLRR ¹⁶	6.2703	7.9171	4.6868	1.5971	0.3316	0.2803
LP ²	6.6600	8.3481	6.8271	1.4904	0.4729	0.3417
DenseFuse ¹⁴	6.2391	7.8940	4.5639	1.5799	0.2912	0.2688
FusionGAN ²²	6.5778	8.7262	3.9830	2.4315	0.1815	0.3030
RFN-Nest ¹⁷	6.8345	8.3471	4.4284	1.6878	0.3604	0.3094
GANMcC ²³	6.8045	8.6032	5.0769	1.9945	0.3159	0.2724
MSPFNet	6.9493	8.5058	6.6112	1.9519	0.5075	0.4009

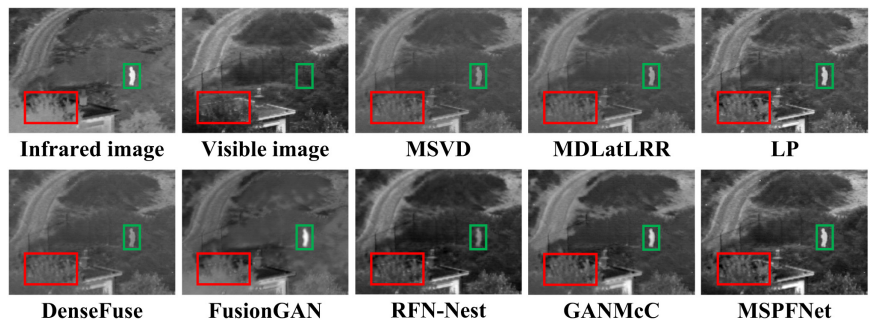


Fig. 6. Qualitative results of MSPFNet and seven comparison methods in an example pair of infrared and visible images from *Nato_camp_sequence* dataset. From top to bottom (from left to right): infrared image, visible image, the fused results of MSVD, MDLatLRR, LP, DenseFuse, FusionGAN, RFN-Nest, GANMcC and our MSPFNet.

and RFN-Nest are weakened. In the marked “tree” regions, MSPFNet has clearer texture details while the texture details of the other methods are distorted.

4.4. RoadScene

Table 3 shows the mean quantitative results of MSPFNet and the seven methods on *RoadScene* dataset. From the table, it can be found that MSPFNet achieves three optimal values (EN, SD, VIF) and two suboptimal values (SF, $Q^{AB/F}$), which illustrates that MSPFNet can preserve more information (EN), and its fused images are with more texture details (SD, SF, $Q^{AB/F}$) and more realistic visual effects (VIF).

Figure 7 shows the qualitative results of MSPFNet and the seven methods in *FLIR_08932* and *FLIR_06325*. Compared with the other seven methods, the fused images of MSPFNet have more prominent targets and richer texture details. As Fig. 7 shows, some local regions in the fused images are marked using red boxes, and they are enlarged for better illustration. Taking image *FLIR_06325* for example, in their marked “cloud” regions, MSPFNet has richer texture details and more

Table 3. Mean quantitative results for fused images on *RoadScene* dataset. (Red: optimal, Blue: suboptimal)

Method	EN	SD	SF	MI	$Q^{AB/F}$	VIF
MSVD ²⁵	6.8435	9.7146	6.2387	2.8147	0.3780	0.3440
MDLatLRR ¹⁶	6.8469	9.7064	5.1916	3.0252	0.4508	0.3613
LP ²	7.1507	9.9151	6.9409	2.6131	0.6152	0.4130
DenseFuse ¹⁴	6.8029	9.6879	4.8336	3.0868	0.3458	0.3429
FusionGAN ²²	7.1005	9.9851	4.7295	2.8818	0.2223	0.2994
RFN-Nest ¹⁷	7.3420	10.1910	4.4965	2.9204	0.2671	0.3728
GANMcC ²³	7.2581	10.2396	5.3031	2.9867	0.3523	0.3544
MSPFNet	7.3458	10.6357	6.7805	2.9356	0.5774	0.4332

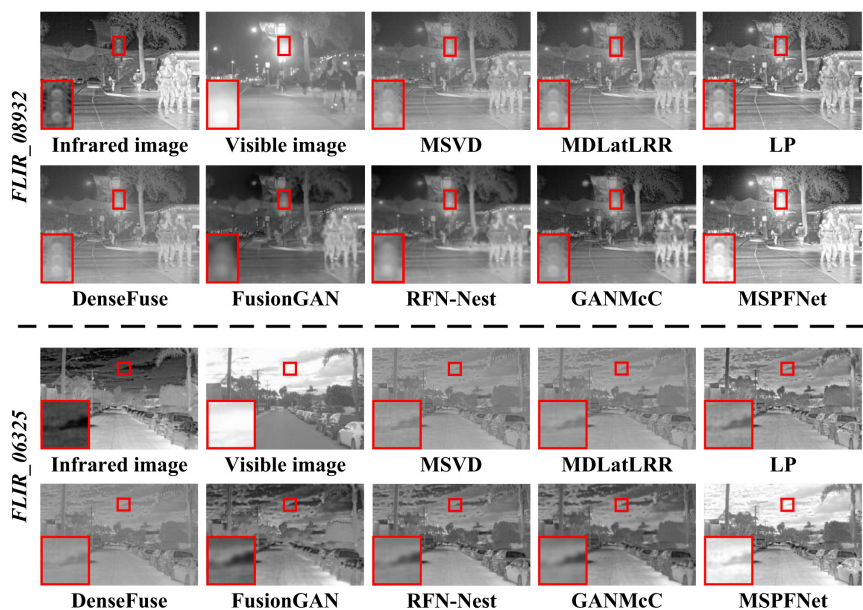


Fig. 7. Qualitative results of MSPFNet and seven comparison methods in two pairs of infrared and visible images from *RoadScene* dataset. From top to bottom (from left to right): infrared image, visible image, the fused results of MSVD, MDLatLRR, LP, DenseFuse, FusionGAN, RFN-Nest, GANMcC and our MSPFNet.

realistic visual effects, while the other methods lose the climate information of the visible image.

4.5. Ablation experiments

4.5.1. Pyramid levels

We evaluate MSPFNet when N varies from 1 to 7.

Table 4 shows the mean quantitative results of MSPFNet with different pyramid levels N on *TNO* and *RoadScene* datasets. Comprehensively, when $N = 1$, MSPFNet performs the worst, which indicates that using the Laplacian pyramid improves the image fusion quality. When $N = 5$, MSPFNet achieves three optimal values, three suboptimal values, one third optimal value, three fourth optimal values, one fifth optimal value and one sixth optimal value on *TNO* dataset and *RoadScene* dataset. In addition, when $N = 5$, by calculating the mean value of the metrics on *TNO* dataset and *RoadScene* dataset, it can be found that MSPFNet achieves two optimal values and two suboptimal values. In other words, when $N = 5$, MSPFNet has the best comprehensive performance compared with other cases.

Figures 8 and 9 show the fused images of MSPFNet for *Kaptein_1654* from *TNO* dataset and *FLIR_00455* from *RoadScene* dataset, when pyramid level N

Table 4. Mean quantitative results of MSPFNet with different pyramid levels N on *TNO* and *RoadScene* datasets. (Red: optimal, Blue: suboptimal)

Dataset	Level	EN	SD	SF	MI	$Q^{AB/F}$	VIF
<i>TNO</i>	$N=1$	5.6142	6.8711	3.6900	2.2181	0.1362	0.1919
	$N=2$	6.8712	8.8665	6.1243	2.3379	0.5678	0.4342
	$N=3$	6.9424	8.8500	6.1505	2.3088	0.5595	0.4428
	$N=4$	7.0171	8.9248	6.1458	2.2847	0.5602	0.4436
	$N=5$	6.9736	8.9833	6.1277	2.4344	0.5522	0.4429
	$N=6$	6.8796	8.8740	6.2064	2.1791	0.5407	0.4250
	$N=7$	6.9340	8.9319	6.1361	2.2200	0.5395	0.4320
<i>RoadScene</i>	$N=1$	6.6454	9.9624	5.0794	3.2810	0.3407	0.2758
	$N=2$	7.3793	10.7160	6.8313	3.0138	0.5769	0.4293
	$N=3$	7.3448	10.6225	6.7948	2.9924	0.5728	0.4272
	$N=4$	7.3616	10.6449	6.8167	2.9143	0.5829	0.4272
	$N=5$	7.3458	10.6357	6.7805	2.9356	0.5774	0.4332
	$N=6$	7.3264	10.7343	6.8453	2.8007	0.5743	0.4248
	$N=7$	7.3382	10.5772	6.8392	2.8118	0.5732	0.4298

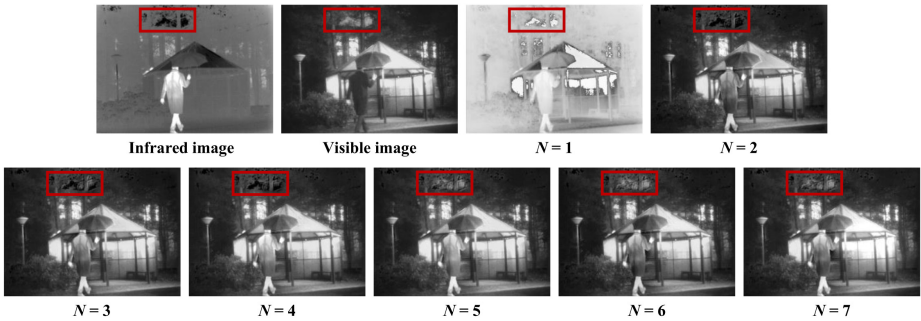


Fig. 8. Qualitative results of MSPFNet with different pyramid levels N in *Kaptein_1654* from *TNO* dataset. From top to bottom (from left to right): infrared image, visible image, the fused results of MSPFNet when $N = 1, 2, \dots, 7$.

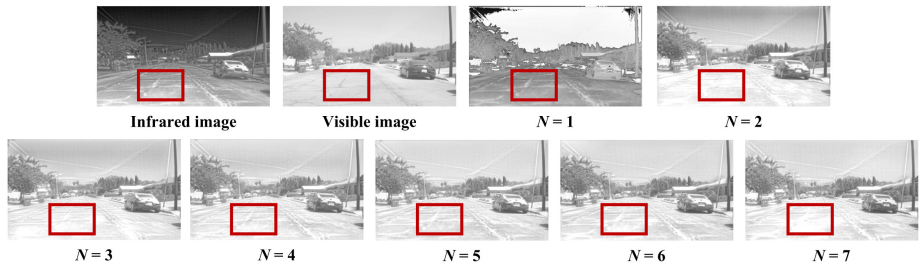


Fig. 9. Qualitative results of MSPFNet with different pyramid levels N in *FLIR_00455* from *RoadScene* dataset. From top to bottom (from left to right): infrared image, visible image, the fused results of MSPFNet when $N = 1, 2, \dots, 7$.

varies from 1 to 7. Taking image *Kaptein_1654* for example, in their marked “sky” regions, MSPFNet cannot describe the natural conditions of the “sky” region when $N < 5$. In Fig. 9, the fused images have more texture details as N increases when $N < 6$. In the marked “ground” regions, it can be seen that the texture details are blurred when $N > 5$. Both in Figs. 8 and 9, the texture details are blurred and the pixels are whitened when $N = 1$, which indicates that using Laplacian pyramid can improve fusion quality. The above experiments show that using the multi-scale information and high-frequency features of source images can help MSPFNet to extract more texture details and improve the visual effects of the fused images.

Based on the above results, we set N as 5.

4.5.2. Channel sum

In this paper, Channel Sum is used to transform the output of each Fusion Subnet. To evaluate its influence, we apply a convolutional layer with output channel of 1 instead of Channel Sum at the end of Fusion Subnet. Figure 10(a) shows the fused images of MSPFNet and Fig. 10(b) shows the fused images of MSPFNet without

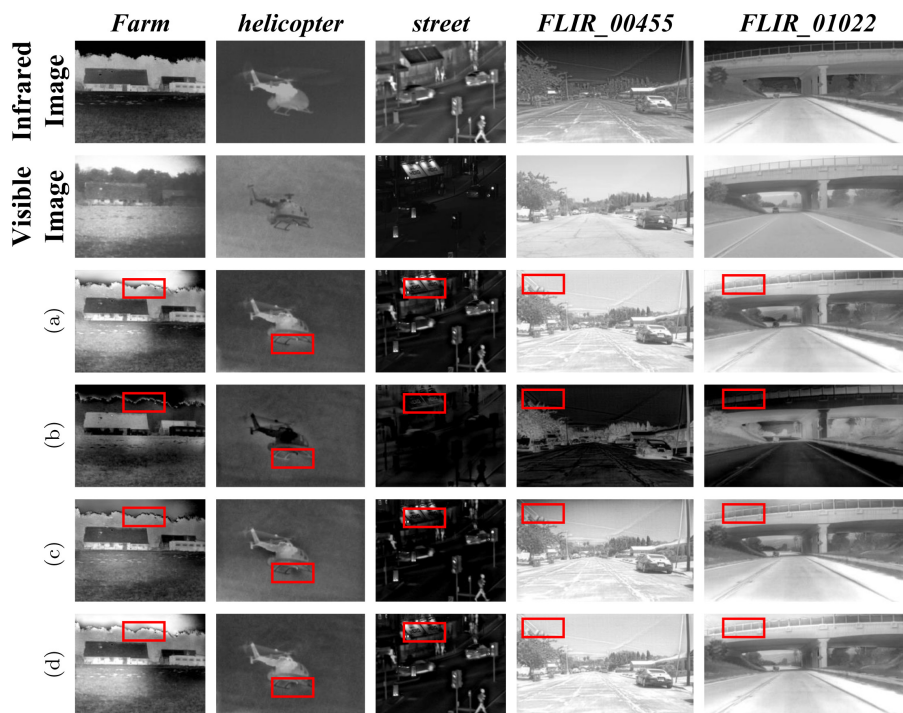


Fig. 10. Qualitative results of MSPFNet and the ablation results of MSPFNet on *TNO* and *Road-Scene* datasets. From top to bottom: infrared image, visible image, (a) fused images of MSPFNet, (b) fused images of MSPFNet without Channel Sum, (c) fused images of MSPFNet with L_{ssg} , (d) fused images of MSPFNet without parameter sharing.

Channel Sum. As Fig. 10(b) shows, without Channel Sum, MSPFNet cannot correctly extract the useful information of source images. Compared with the whole MSPFNet, MSPFNet without Channel Sum loses the texture details and intensity information, which indicates that Channel Sum can help MSPFNet to preserve more target intensity information in infrared images and more texture details in visible images.

4.5.3. Multi-scale gradient loss

By comparing the multi-scale difference among Gaussian pyramid components of two source images and that of the fused image, we design the multi-scale gradient loss to train MSPFNet. To evaluate its influence, we reset the loss function as Eq. (4.1). Different from L_{msg} , L_{ssg} only compares the first layer gradient difference among Gaussian pyramid components of two source images and that of the fused image:

$$L_{\text{ssg}} = \|\nabla G_1^*(I_f) - \max(\nabla G_1(I_r), \nabla G_1(I_v))\|_1. \tag{4.1}$$

Figure 10(c) shows the fused images of MSPFNet with L_{ssg} . Compared with Fig. 10(a), the fused images have more edge artifacts, and texture details are more blurred in Fig. 10(c). In image *Farm*, MSPFNet with L_{ssg} cannot extract the realistic climate conditions in the “sky” region. As Fig. 10(c) shows, MSPFNet with L_{ssg} produces obvious artifacts. In image *street*, the characters on the advertising board are blurred using L_{ssg} instead of L_{msg} . To sum up, multi-scale gradient loss can help MSPFNet to extract richer texture details, and the fused images are with more realistic visual effects.

4.5.4. Parameter sharing for fusion subnets

In order to reduce the network complexity, we share parameters among the Fusion Subnets. To evaluate its influence, we do not share parameters among layers of the pyramid to get the fused results. Figure 10(d) shows the fused results of MSPFNet without parameter sharing. As shown in Figs. 10(a) and 10(d), the fused results of MSPFNet with and without parameter sharing are similar. Table 5 shows the mean quantitative results of MSPFNet with and without parameter sharing on *TNO* and *RoadScene* datasets.

Table 5. Mean quantitative results of MSPFNet with and without parameter sharing on *TNO* and *RoadScene* datasets.

Method	Dataset	EN	SD	SF	MI	$Q^{AB/F}$	VIF
MSPFNet with parameter sharing	<i>TNO</i>	6.9736	8.9833	6.1277	2.4344	0.5522	0.4429
	<i>RoadScene</i>	7.3458	10.6357	6.7805	2.9356	0.5774	0.4332
MSPFNet without parameter sharing	<i>TNO</i>	6.9667	8.9072	6.0897	2.2958	0.5678	0.4379
	<i>RoadScene</i>	7.3522	10.7436	6.8275	2.9012	0.5916	0.4297



Fig. 11. Qualitative results of MSPFNet in two pairs of infrared and visible images from *RoadScene* dataset. From left to right: infrared image (Gray), visible image (RGB) and the fused results of MSPFNet (color online).

RoadScene datasets. The optimal values on the corresponding dataset are highlighted in bold font. Comprehensively, on *TNO* dataset, MSPFNet with parameter sharing performs better. On *RoadScene* dataset, MSPFNet without parameter sharing performs better. However, the model size of MSPFNet with parameter sharing (0.1831M) is much smaller than that of MSPFNet without parameter sharing (0.9154M). So we share parameters among the Fusion Subnets.

4.6. Fused results for infrared and RGB image

Figure 11 shows the qualitative results of MSPFNet in infrared and RGB visible images on *RoadScene* dataset. The visible image is first transformed into YCrCb channels. Then, using MSPFNet, the fusion process is produced on the component of Y channel of the visible image and the infrared image. Combining the Cr and Cb channels of the visible image, the fused result is finally transformed into RGB channels as Y channel. As Fig. 11 shows, MSPFNet can capture intensity and contour information of infrared images and obtain the color information and texture details of visible images.

4.7. Running time comparison

Table 6 shows the average running time of different deep learning-based methods on the *TNO*, *Nato_camp_sequence* and *RoadScene* datasets. These methods are tested

Table 6. Average running time of our MSPFNet and four deep learning-based methods on *TNO*, *Nato_camp_sequence* and *RoadScene* datasets. (Red: optimal, Blue: suboptimal; Unit: second)

Dataset	DenseFuse	FusionGAN	RFN-Nest	GANMcC	MSPFNet
<i>TNO</i>	0.1302	0.7387	0.2090	1.2615	0.1353
<i>Nato_camp_sequence</i>	0.0605	0.1075	0.0886	0.1320	0.0748
<i>RoadScene</i>	0.0780	0.6919	0.1397	1.0200	0.0995

on an NVIDIA GeForce RTX 2080Ti GPU. MSPFNet performs the second fastest out of all comparison methods. Although DenseFuse has a shorter running time than MSPFNet, MSPFNet preserves richer texture details and has more realistic visual effects.

5. Conclusion

We propose a multi-scale pyramid network (MSPFNet) for infrared and visible image fusion by combining deep learning and image Laplacian pyramid. MSPFNet does not require hand-crafted fusion rules and can efficiently extract and fuse the multi-scale information. Qualitative and quantitative experimental results on publicly available datasets *TNO* and *RoadScene* show that MSPFNet can extract more texture details and have more realistic visual effects on the fused images. In future work, we will explore the possibility of combining deep learning methods with other traditional methods to improve the quality of infrared and visible image fusion.

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